

CONSTRUCTION TECHNOLOGY

Construction technology has always played a central role in society. It is a multifaceted industry in which projects can result in products, systems, processes, or services and encompass various aspects of production, repair, and maintenance. In construction technology courses, students will learn about designing, constructing, and maintaining a variety of buildings and structures, and will gain experience with the tools, equipment, and processes commonly used in the field. Students will also learn about health and safety standards in the construction industry, building codes and regulations, and employment opportunities and careers in the various sectors of the industry, including electricity, carpentry, masonry, heating and cooling, and plumbing.

The construction technology courses initially focus on residential and light construction systems related to residential buildings and structures, and progress to more advanced residential and light commercial construction. The woodworking courses allow students to concentrate on furniture building or cabinet making.

The list of approved emphasis areas for construction technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Construction Technology, Grade 12

Workplace Preparation

TCJ4E

This course enables students to further develop technical knowledge and skills related to residential construction and to explore light commercial construction. Students will continue to gain hands-on experience using a variety of materials, processes, tools, and equipment; create and interpret construction drawings; and learn more about building design and project planning. They will expand their knowledge of terminology, codes and regulations, and health and safety standards related to residential and light commercial construction. Students will also expand their awareness of environmental and societal issues related to construction technology, and will explore entrepreneurship and career opportunities in the industry that may be pursued directly after graduation.

Prerequisite: Construction Technology, Grade 11, Workplace Preparation

A. CONSTRUCTION TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** assess the properties and applications of a variety of construction materials, tools, equipment, and processes;
- A2.** demonstrate an understanding of building codes, regulations, and standards for construction projects;
- A3.** use construction terminology correctly;
- A4.** apply mathematical skills and scientific concepts in the planning and building of a variety of construction projects.

SPECIFIC EXPECTATIONS

A1. Materials, Tools, Equipment, and Processes

By the end of this course, students will:

- A1.1** determine appropriate building materials, tools, and equipment for various construction projects (*e.g., materials: wood products, stone, cultured stone, steel studs, copper wire, cast-iron pipe and fittings, polyvinyl chloride [PVC] pipe and fittings, membranes, adhesives; tools: saws, power nailers; equipment: cement mixer, air compressor, generator, pressure gauge, CO meter*);
- A1.2** compare the properties (*e.g., mechanical, structural, thermal*) of natural and manufactured building materials;
- A1.3** compare the advantages and disadvantages of various structural and non-structural materials in construction projects (*e.g., wood versus steel for studs and beams; asphalt shingles versus steel roofing; cement block versus poured concrete; brick versus vinyl, aluminum, or wood siding*);
- A1.4** describe various processes in residential and/or light commercial construction (*e.g., assembling platform framing, pouring concrete, laying brick or block, setting tile, installing structural or reinforcing steel*).

A2. Codes, Regulations, and Standards

By the end of this course, students will:

- A2.1** correctly identify and interpret the sections of building, electrical, and fire codes that apply

to residential and/or light commercial construction projects in Ontario (*e.g., Ontario Building Code sections for built-up beams, mechanical fasteners, renovations, and potable-water, storm-water, and drain/waste/vent systems; Ontario Electrical Safety Code chapters for residential and/or commercial electrical wiring and equipment; Ontario Fire Code regulations for fire barriers, air flow, sprinklers, smoke detectors, and fire alarms*);

- A2.2** describe other regulations that apply to construction projects (*e.g., municipal by-laws, conservation authority restrictions, requirements for barrier-free access*);

- A2.3** describe safety standards and regulations that apply to construction projects (*e.g., standards and/or regulations from the Canadian Standards Association [CSA], Underwriters Laboratories of Canada [ULC], the Workplace Safety and Insurance Board [WSIB], the Technical Standards and Safety Authority, and the Occupational Health and Safety Act*).

A3. Terminology

By the end of this course, students will:

- A3.1** use correct terminology for materials (*e.g., girders, open-web steel joists, plenums, medium-density fibreboard, millwork, bricks, mortar*);
- A3.2** use correct terminology for components and concepts (*e.g., components: flying form, vent stack, floor drain, plumbing trap, ground-fault circuit interrupter; concepts: wet vent, superheat, flashgas*).

control-circuit hunting, subcooling, safety factor, shear force, tensile strength);

A3.3 use correct terminology to identify and describe construction tools and equipment (e.g., *tools: builder's level, framing hammer, wire stripper, pliers, tri-square, trowel, pipe cutter, hand saw, reciprocating saw, masonry saw, circular saw, drill; equipment: air compressor, scaffolding, cement mixer, generator, electrical test meter*);

A3.4 use correct terminology to describe types of residential and/or light commercial buildings (e.g., *residential: bungalow, two-storey, townhouse, side-split; commercial: office, motel, warehouse, mercantile*);

A3.5 use correct terminology to describe common engineering features of buildings (e.g., *engineered floor joists, roof trusses, structural steel, fire barriers*).

A4. Mathematical Skills and Scientific Concepts

By the end of this course, students will:

A4.1 convert between fractions and decimals and between imperial and metric units (e.g., *between feet and metres, between Fahrenheit and Celsius temperatures*), using appropriate tables, charts, software, and/or online conversion tools;

A4.2 use mathematical concepts and formulas (e.g., *trigonometric functions, Pythagorean theorem*) to lay out construction projects;

A4.3 use the tables in Part 9 of the Ontario Building Code to determine the required sizes, spacings, and numbers of structural members (e.g., *footings, concrete blocks, floor joists, wall studs, rafters, beams, columns*);

A4.4 prepare detailed estimates of quantities and costs of materials for construction projects (e.g., *concrete, lumber, roofing, millwork, finishes*), and of labour costs for these projects;

A4.5 perform calculations related to the installation of various building systems (e.g., *electrical: calculate conduit fill for conductors of various sizes and types, power in two- and three-wire systems, and required conductor size and overcurrent protection for various loads; plumbing: calculate flow rates, 45° and parallel offsets for piping, required size and slopes for drains; carpentry: calculate angles and rafter sizes for pitched roofs, and riser, tread, and floor-opening sizes for stairs; heating and cooling: calculate output of HVAC systems and heat gain and loss of buildings*);

A4.6 demonstrate an understanding of scientific concepts related to construction technology (e.g., *relationships among voltage, current, resistance, and power, and between wire size and current capacity; simple electron theory; methods of heat transfer*).

B. DESIGN, LAYOUT, AND PLANNING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply a design process and/or other problem-solving methods as appropriate when planning a variety of residential and/or light construction projects, and demonstrate an understanding of the design considerations for these projects;
- B2.** create and interpret working drawings for residential and/or light commercial construction projects;
- B3.** apply accurate technical data and relevant building codes, regulations, and standards when planning and developing construction projects;
- B4.** plan and lay out systems for residential and/or light commercial buildings.

SPECIFIC EXPECTATIONS

B1. Design and Planning

By the end of this course, students will:

- B1.1** apply the steps of a design process and/or other problem-solving techniques (see pp. 21–23) to solve a variety of construction technology challenges or problems;
- B1.2** identify the factors that affect site layout and preparation, and allow for these factors when planning and designing construction projects;
- B1.3** compare design considerations for a residential property with those for a light commercial property (*e.g., client needs and budget; location, type, and use of building; connection to utilities; zoning restrictions, codes, regulations, and standards; barrier-free access; aesthetics; energy efficiency*);
- B1.4** explain how the design process is used in the construction industry (*e.g., with respect to precast concrete, prefabricated components, interior and exterior finishes*);
- B1.5** apply design principles when designing aspects of construction projects (*e.g., kitchen, electrical, mechanical, and plumbing layout; means of egress; customer traffic flow; display space*), and describe how various systems are integrated into the overall design;
- B1.6** create and/or modify the design of a system for a construction project (*e.g., water supply and drains, roof structure and membrane, furnace and ductwork, control circuit using relay logic and/or a programmable logic controller*).

B2. Working Drawings

By the end of this course, students will:

- B2.1** produce sketches and/or working drawings with manual and/or computer-assisted methods (*e.g., electrical riser diagram, stack elevation drawing, wall section*), using appropriate metric and/or imperial units;
- B2.2** use working drawings to help plan the sequence of tasks for completing a construction project (*e.g., excavation, footings, foundation walls, framing, sheathing, rough-in of mechanical systems, drywalling, installation of devices and fixtures*);
- B2.3** interpret working drawings to accurately lay out construction projects (*e.g., determine dimensions and placement of footings, columns, openings, and beams; determine specified materials, size and type of windows and doors, masonry bond pattern, and locations of electrical, mechanical, and plumbing components*).

B3. Codes, Regulations, and Standards

By the end of this course, students will:

- B3.1** outline the building code requirements for forces and stresses due to loads and weather conditions (*e.g., heat, cold, wind, snow, rain*) that structural members (*e.g., footings, foundations, floors, bearing walls, roofs, columns, beams, lintels*) must be designed to withstand;

B3.2 use various resources to find technical data, code requirements, and standards for construction projects (*e.g., reference charts and tables; reports; published codes, regulations, and standards; guides and trade manuals; manufacturers' instructions; government and association websites*);

B3.3 describe the permit and inspection process required for residential and/or light commercial construction projects (*e.g., for demolition, building, electrical system, plumbing, heating, cooling*).

B4. Building Systems

By the end of this course, students will:

B4.1 identify and describe the factors that affect the design and installation of foundations for construction projects (*e.g., drainage, soil conditions, frost penetration, reinforcement, insulation*), and allow for these factors when planning and designing a construction project;

B4.2 describe and plan the relationship between structural and decorative elements in construction (*e.g., studs, toe plates, and top plates provide secure support for baseboards and mouldings, which cover joints where walls meet floors and ceilings; masonry bond patterns provide strength and attractive appearance*);

B4.3 research, identify, and recommend modifications to a building project that would improve its quality and value (*e.g., upgrades to insulation, air barrier, or HVAC system that reduce operating costs; use of more attractive or durable fixtures, trim, flooring, or exterior finish*).

C. FABRICATION, ASSEMBLY, AND FINISHING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** apply appropriate technical skills, including the safe use of the tools, equipment, and materials required to build construction projects;
- C2.** apply safe and accurate techniques for building construction projects;
- C3.** complete construction projects by correctly applying finishing materials and installing fixtures and devices.

SPECIFIC EXPECTATIONS

C1. Technical Skills

By the end of this course, students will:

- C1.1** use safe work practices with all construction tools, materials, and equipment;
- C1.2** lay out construction projects using a variety of tools and equipment (*e.g., framing square, laser level, string line, plumb bob*);
- C1.3** install various systems of residential and/or light commercial construction projects (*e.g., structural, electrical, plumbing, masonry, heating/ventilation/air-conditioning*) safely and in accordance with codes, regulations, and standards;
- C1.4** use metric and imperial units correctly and accurately during the construction of various building systems;
- C1.5** use various methods to communicate construction information clearly (*e.g., written instructions, sketches, detail drawings*).

C2. Building and Assembly

By the end of this course, students will:

- C2.1** safely construct residential and/or light commercial projects in accordance with design specifications (*e.g., architect's drawings; engineering specifications; fixtures, trim, cabinetry, flooring, or paint chosen by client or interior designer; manufacturers' installation instructions*);
- C2.2** use appropriate solutions for problems in the construction process, and recognize when modifications require a change order.

C3. Finishing

By the end of this course, students will:

- C3.1** apply appropriate exterior materials to finish residential and/or light commercial construction projects (*e.g., curtain walls, foundation coatings, stucco, brick veneer, metal siding*);
- C3.2** apply appropriate interior materials to finish residential and/or light commercial construction projects (*e.g., suspended ceiling, baseboards, window and door casings, mouldings, built-ins, ceramic tiles*).

D. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of the environmental effects of construction projects, and ways of reducing harmful effects;
- D2.** demonstrate an understanding of how the construction industry affects society.

SPECIFIC EXPECTATIONS

D1. Technology and the Environment

By the end of this course, students will:

- D1.1** assess environmentally friendly alternatives for building systems (*e.g., heating with solar energy, heat pumps, or geothermal systems; reusing grey water; harvesting rainwater; chlorine-free treatment of storm water and sewage*);
- D1.2** assess the environmental and health effects of using manufactured construction materials (*e.g., pressure-treated wood, oriented-strand board, medium-density fibreboard, cultured stone*);
- D1.3** compare ways of reducing and/or managing energy consumption in homes and businesses (*e.g., smart meters, timers, skylights, heat recovery, energy-efficient lighting*);
- D1.4** research and describe strategies for implementing sustainable building practices (*e.g., Canada Green Building Council guidelines, Forest Stewardship Council Canada standards*).

D2. Technology and Society

By the end of this course, students will:

- D2.1** identify the economic and social effects of the construction industry (*e.g., waste disposal, land use, labour supply and cost, water supply, local infrastructure*);
- D2.2** identify factors to consider in community planning (*e.g., population density, culture, the environment*);
- D2.3** assess the societal and cultural impact of construction-related emerging technologies from various perspectives (*e.g., Aboriginal, safety, technical, financial, business*).

E. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** demonstrate an understanding of and comply with health and safety regulations and practices specific to the construction industry;
- E2.** describe the organization and management of construction companies and the entrepreneurial skills needed to establish a successful construction business;
- E3.** describe the skills and training required for careers in the construction industry.

SPECIFIC EXPECTATIONS

E1. Health and Safety

By the end of this course, students will:

- E1.1** describe hazards related to construction materials, processes, and equipment (*e.g., toxic or flammable fumes from solvents, paints, varnishes, and gasoline; explosion or burns from propane; lung damage from silica; tripping or falls in unfinished buildings; shock from damaged power tools or electrical equipment*), and the precautions that should be taken to avoid these hazards;
- E1.2** describe and comply with health and safety legislation and practices for the construction industry (*e.g., Workplace Safety and Insurance Board [WSIB] regulations, provincial labour legislation, the Ontario Building Code, local by-laws*);
- E1.3** use, handle, and store materials in accordance with Workplace Hazardous Materials Information System (WHMIS) guidelines;
- E1.4** describe and follow proper procedures for locking out equipment, and for the set-up, use, and maintenance of trenches, ladders, and scaffolding in accordance with Occupational Health and Safety Act regulations and Construction Safety Association of Ontario guidelines;
- E1.5** describe the rights and responsibilities of employees (*e.g., the right to know, the right to refuse, the right to participate, as outlined in the Occupational Health and Safety Act*);
- E1.6** demonstrate an understanding of when and how to use appropriate protective clothing, gear, and equipment (*e.g., hard hat, respirator, safety harness*).

E2. Construction Business

By the end of this course, students will:

- E2.1** describe the organization and management of a typical residential and/or commercial construction company (*e.g., roles of general contractor, builder, subcontractor, renovator, lead hand, shop steward, superintendent, and site manager*);
- E2.2** identify documents commonly used in the planning and management of the construction process (*e.g., schedules, contracts, Gantt charts, permits, subtrade contracts*);
- E2.3** identify the factors to be considered when starting a construction company (*e.g., potential clients, product offered, competition, start-up costs, business plan*);
- E2.4** describe the skills important for success as an entrepreneur (*e.g., creativity, leadership, problem solving, time management, self-motivation, organization*).

E3. Career Opportunities

By the end of this course, students will:

- E3.1** describe career opportunities in the construction industry (*e.g., general contractor, builder, tradesperson, technician, technologist, labourer, estimator, engineer, interior decorator, artisan*);
- E3.2** describe the education and training required for careers in the construction industry (*e.g., apprenticeship, postsecondary courses*), and list the entrance requirements for these programs;

E3.3 demonstrate an understanding of and apply the Essential Skills that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., decision making, document use, measurement and calculation*);

E3.4 demonstrate an understanding of and apply the work habits that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, initiative*);

E3.5 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in construction technology (*e.g., Passport to Safety certificate, technical drawings, reports, photographs of projects, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.